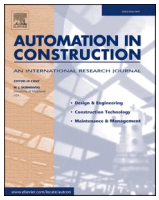
Automation in Construction 139 (2022) 104312



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





Vision-based monitoring of site safety compliance based on worker *re*-identification and personal protective equipment classification

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ARTICLE INFO

Keywords:
Computer vision
Deep learning
Personal protective equipment classification
Safety compliance monitoring
Worker *re*-identification

ABSTRACT

Construction sites are highly hazardous due to the dynamic interaction between workers and moving equipment, with high fatality rates caused by collision and falling from height, etc. Hence, identifying unsafe behaviors among workers is crucial for enhancing site safety, such as tracking their on-site movement and personal protective equipment (PPE). Vision-based video processing has been actively used to automatically recognize workers and their behaviors on construction sites. However, existing studies mainly monitor workers within a single camera capturing only a small sub-region. As workers typically move around fairly large sites, continuously tracking their movement across multiple cameras would enable more comprehensive behavioral analyses. Hence, this paper proposes a framework for monitoring safety compliance among workers, by combining worker re-identification (ReID) and PPE classification. Deep learning-based approaches are developed to address the challenges for these two tasks respectively. For ReID, a new loss function named similarity loss is designed to encourage deep learning models to learn more discriminative human features, realizing a more robust tracking of individual workers. For classifying PPE statuses, a weighted-class strategy is proposed to mitigate model bias when given imbalanced samples among classes, for improved performance despite limited training samples. By combining the ReID and PPE classification results, a workflow is developed to log any incident of workers not wearing the necessary PPEs. With an actual construction site dataset, the proposed methods improve worker ReID and PPE classification by 4% and 13% accuracies respectively, which will facilitate site video analytics and inspection of site safety compliance among workers.

1. Introduction

The construction industry is one of the very hazardous industries, with high accident rates recorded in the past years. From 2013 to 2019, the construction industry in Hong Kong had the highest fatality rates every year among the 14 main industry sectors [1,2]. The United States also reported that in 2019 about 20% of occupational fatalities occurred on construction sites [3]. On-site accidents have led to significant financial losses among stakeholders [4], e.g. due to compensation claims, which necessitate systematic practices of site safety monitoring. Construction sites involve highly dynamic interactions between multiple workers and equipment (e.g. excavators, trucks) [5]. In particular, the major causes for the construction injuries and fatalities include collision with moving objects and falling from height [2]. To avoid severe injuries, any unsafe behavior among workers should be identified and rectified. Examples include analyzing the on-site movements of workers and inspecting the proper wearing of personal protective equipment (PPE), whose records can be logged into regular site diaries for review and behavioral analyses. The traditional practices of site safety monitoring mainly involve on-site safety officers for manually observing the behavior and job operations of workers [6]. Yet, continuously monitoring workers scattered over a large site is labor-intensive and error-prone, where unsafe behavior may be overlooked.

To this end, automated processing of surveillance camera videos can potentially facilitate site monitoring. Surveillance cameras have been commonly deployed for various purposes, e.g. identifying criminal suspects in airports [7] and monitoring construction site activities [8,9]. By incorporating computer vision and deep learning techniques into surveillance systems, the visual content on video frames can be automatically interpreted. Taking supervised learning as an example, a deep convolutional neural network (CNN) is trained with images whose desired output is manually labeled, such that the network learns to

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https://doi.org/10.1016/j.autcon.2022.104312
Received 6 October 2021; Received in revised form 16 April 2022; Accepted 29 April 2022
Available online 6 May 2022
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extract useful features from images for various tasks, e.g. object detection and classification [10,11]. These techniques have also been applied in civil engineering [12], e.g. defect detection for structural health assessment [13,14], monitoring of construction sites such as location tracking and activity classification of workers and equipment on sites, site accident monitoring, and productivity evaluation [15,16]. Some studies also focused on identifying whether workers wore the necessary PPEs, e.g. safety helmets and reflective vests [17,18]. Nevertheless, existing studies only monitored individual workers within a small sub-region captured by a single camera, i.e. the movement of workers cannot be continuously tracked over the whole site. Since workers usually move around different areas of a site, continuous wide-area tracking of workers would enable more comprehensive behavioral analyses and safety monitoring.

Hence, this paper focuses on worker tracking across multiple cameras with non-overlapped fields-of-view, in order to continuously trace individual workers over the whole site. In particular, such across-camera tracking can be formulated into a person re-identification (ReID) problem [19]: Upon collecting raw videos from multiple cameras, individuals are first detected by bounding boxes. Then, discriminative human features are extracted from each detected person, in order to achieve accurate identity matching, i.e. assigning a unique and consistent identity to all images of the same person. While across-camera ReID is rarely incorporated into the monitoring of large-area movement among construction workers, it can potentially enable more comprehensive safety compliance monitoring. In particular, using PPE classification techniques can identify the necessary PPEs on each worker image, whereas ReID techniques can assign a consistent identity to each image. By combining these two sources of results, whether individual worker is wearing the necessary PPEs can be continuously monitored.

Nevertheless, person ReID and PPE classification respectively face several challenges as revealed in recent deep learning-based studies. For ReID, human appearance across different cameras is subject to intra-person variation and inter-person ambiguity [19,20], making identity matching inaccurate if discriminative features are not exploited. For classifying different PPEs, existing studies mostly overlook the problem of imbalanced training samples among PPE classes [18], i.e. only low proportions of training samples belong to some minority classes, for which a model may hardly learn relevant features from limited samples [21]. Solving this problem is practically crucial since unsafe behaviors violate on-site safety rules, such that training images of missing-PPE classes are scarce, constituting the minority classes which require competitive performance with limited samples.

To address the aforementioned challenges, this paper proposes an integrated framework, which combines worker ReID and PPE classification based on deep learning for more comprehensive site safety monitoring. More effective methodologies are developed for each process to improve their performances over existing methods. For ReID, a new loss function named similarity loss is designed for training deep learning models, which further facilitates the learning of discriminative human features for more robust worker tracking across multiple cameras. For PPE classification, a weighted-class strategy is incorporated into a training loss function for addressing imbalanced samples among classes, where the model bias towards majority classes is effectively mitigated, i.e. improving model performance despite limited training samples. Eventually, a workflow that combines the ReID and PPE classification results is developed, for counting the incidents of workers missing the necessary PPEs. The remainder of this paper covers related research (Section 2), the developed methodology (Section 3), experimental results on an actual construction site (Section 4), conclusions and future work (Section 5).

2. Related works

2.1. Computer vision for construction monitoring

Video analytics equipped with computer vision techniques have been widely applied in civil infrastructure monitoring. Related applications include the assessment of structural damage conditions for bridges, buildings or underground utilities [14,22–24]. For example, Dung et al. [23] used a CNN to classify whether cracks exist around the welded joints of steel bridges, automatically from hand-held camera images. Wang and Cheng [14] developed a unified CNN for geometric and semantic segmentation of underground pipe defects, with images captured during site inspection. Xiong et al. [24] instead analyzed aerial images taken by unmanned drones capturing a group of buildings, for identifying the collapsed buildings after an earthquake. These studies mostly collected images from mobile cameras, which requires tedious planning of the image-taking process to capture the structural elements of interest. Hence, other studies processed videos from stationary surveillance cameras, e.g. for urban traffic analytics and road design. For example, traffic volumes were estimated for optimizing the timing design of signalized junctions, based on automatic movement tracking of pedestrians [25] and vehicles [26]. By pedestrian tracking, Wong et al. [27] analyzed the movement patterns e.g. speed-density relations of different pedestrian groups, by recognizing the trajectory distribution and human attributes of urban crowds.

Besides urban scenarios, video analytics have also been applied to monitoring construction site objects including workers and equipment, e.g. location tracking [28–30] and activity classification [5,15,16,31,32]. The extracted geometric and semantic information then enable site safety assessment. For example, Hongjo et al. [15] tracked the movement of site objects in order to issue real-time proximity warnings before workers were struck by nearby equipment. By movement tracking, Shuai et al. [16] identified workers entering predefined danger zones which instantaneously triggered a warning. In addition, evaluation of the productivity of site operations can also be facilitated by analyzing the object movement and activities. For instance, Dominic et al. [32] explicitly identified the key body joints of each worker (i.e. body posture), whose activities were recognized and hence the duration of performing different tasks was counted for productivity evaluation. Monitoring the movement and activities of construction workers on sites is crucial for behavioral analyses.

Existing studies have a critical limitation, i.e. the worker movement is monitored only within a small sub-region of the whole construction site. More specifically, previous research in worker tracking mostly maintained their identities within a single camera, such that the analyzed walking trajectories were limited to a small region. However, workers may walk over a fairly large site, where continuous tracking of their movement over a wide area would enable a more comprehensive behavioral analyses among workers. For example, safety officers can inspect whether each worker has worn necessary PPEs continuously across different working areas on a large site. Such wide-area monitoring is difficult using a single camera, whose limited field-of-view only covers a local area. Hence, worker tracking across multiple cameras is a solution to more comprehensive and wider-scale behavioral analyses among workers. To this end, worker identities should be accurately matched across multiple cameras, even a worker disappears for a while and is detected again later, e.g. leaving and entering different camera views arbitrarily, or being occluded by background obstacles. Compared with overlapped camera views, ReID among non-overlapped videos covers a wider monitoring area, which however is more challenging because of the uncertain movement and appearance of individual workers across different cameras. Hence, a robust methodology is needed for worker tracking across multiple cameras.

2.2. Person ReID across multiple cameras

Generally, tracking multiple persons across multiple cameras can be formulated into a person ReID problem. More detailed elaborations of the ReID procedures were provided by Wu et al. [19] and Ye et al. [20]. Given videos from different cameras with non-overlapping fields-of-view, the captured persons in each frame are first detected, e.g. as a bounding box cropped from the raw image. The bounding boxes are then processed by a ReID model which extracts discriminative human features from each person, and the feature similarities are then compared to match the same identity while differentiating different identities. Hence, ReID aims at consistent identities across multiple cameras.

To this end, previous studies of ReID encountered several technical challenges [19,20], which can be categorized into two aspects: 1) *intra-person appearance variation*. Since the same person is captured in different locations, the appearance may largely vary due to changes of the view angle, clothing, lighting conditions or partial occlusion of the human body. Identities may be lost if the extracted features are sensitive to such variations. 2) *inter-person appearance ambiguity*. Since an arbitrary group of persons are captured, they may have very similar visual features, such as clothing color, style and certain belongings. Identities may be confused if relying on some generic feature descriptors that are not distinctive enough. Therefore, these two challenging conditions necessitate robust ReID methods which can extract identity-discriminative visual features.

2.2.1. Feature extraction models

Traditional ReID studies relied on hand-crafted human features, e.g. color histograms [33–35], histograms of oriented gradients (HOG) [36] and point-of-interest features like SURF [37]. Zheng et al. [38] provided a more comprehensive survey of the traditional methods for ReID. The traditional methods extract pre-designed feature representations, which are not rich enough for robust identity matching and differentiation. Hence, deep learning has been widely adopted to enable a data-driven process of feature learning, i.e. a model automatically exploits what features are useful and optimizes the feature representations learned. In particular, the state-of-the-art ReID methods mostly consist of CNN architectures [19,20]. Existing ReID studies have designed different CNN variants for extracting discriminative features from images.

Recently published ReID models typically used CNN architectures such as VGG [39] and ResNet [11] as the backbone, upon which different ReID-specific mechanisms were incorporated. For example, a model can be guided to learn localized features by partitioning a human body for part-by-part feature extraction [40,41], or based on human keypoints detected for extracting features from each semantic part, e.g. head and body [42,43]. Moreover, learnable attention modules were designed to exploit discriminative features from the most informative body regions, by weighting the feature maps by their importance among the pixels and channels [44,45]. Furthermore, multi-scale features were extracted by different sizes of receptive fields [46,47], or aggregated among multiple CNN branches which extract features from coarse to fine scales [48]. Wei et al. [49] extracted multi-scale spatial features from each of the consecutive frames of the same person, which were aggregated by a recurrent neural network to produce more robust temporal features. This is one of the few studies in worker ReID, whose experimental evaluation was not detailed enough. For example, they did not evaluate the generalizability of the ReID model to different scenes and its adaptability with different CNN backbones, which will benefit future research in optimizing and deploying the ReID method.

2.2.2. Training loss functions

In addition to CNN architectures, different loss functions have also been designed to facilitate the learning of feature representation. In particular, the training of a ReID model is typically formulated as a closed-set classification problem [50], i.e. learning the features that assist in classifying the identity label of each person. Previous ReID studies [50–52] mostly adopted the softmax loss as the basic setup, which was originally designed for object classification. Given the training images, the final fully connected layer learns to map the extracted feature vectors to a fixed set of identity labels. The learning effect of the softmax loss is to enlarge the inter-class feature separation, i.e. the centroids of feature vectors belonging to different identities are maximized [53], in order to achieve an unambiguous classification of the identity labels.

However, the softmax loss teaches a model to classify a known number of person IDs, whose training objective is misaligned with the testing process of ReID [50]. When actually deployed for ReID, the number of unique identities is unknown beforehand, hence the formulation is no longer classification-based. Instead, only the feature vectors extracted by a model are processed for comparing visual similarities. Consequently, the last fully connected layer is typically discarded in the testing stage. This implies that redundant model parameters are involved in the existing classification-based training regime [54], which may not effectively encourage the feature representation learning of a ReID model [52]. Furthermore, such training regime only coarsely optimizes the feature representations in an inter-class scale, while the intra-class feature similarities among images of the same person are not exploited. This may lead to a dilemma that an inter-class image pair is more similar than an intra-class pair [55], possibly degrading the identity matching accuracy.

To enable a finer scale of feature learning, other losses can be incorporated to explicitly compare the features among image pairs. Hermans et al. [54] designed a strategy of hard triplet sample mining, i.e. hard triplet loss, which enforces a more direct supervision on the feature representation being trained. Given a batch of training images, the hard triplet loss aims to make the features between the positive identity pairs more similar than that between the negative pairs. Consequently, the intra-class feature vectors are made relatively more compact, i.e. the model learns a feature representation that captures the visual similarities among images of the same person. The two existing losses were jointly utilized by Wang et al. [48], with the softmax loss serving a classification-based training regime and the hard triplet loss for supervising the pair-wise feature learning. However, the hard triplet loss compares the feature vectors in a fairly local scale, i.e. only iteratively comparing the pair-wise feature similarities rather than correlating the features among the whole set of training images [56]. Overall, the existing ReID losses may not guarantee effective learning of feature representations, for which a more robust loss function is needed.

2.3. Worker PPE classification

In addition to walking trajectories, the wearing of necessary PPEs should also be monitored for analyzing the safety compliance of workers. Recent vision-based studies of PPE classification mostly localized each worker from a video frame as a bounding box, from which the PPE status of the worker was classified. For example, Rubaiyat et al. [57] classified whether a worker was wearing safety helmet by localizing several helmet colors, followed by shape template matching for confirming the existence of helmets on an image. Mneymneh et al. [58] detected candidate regions of a helmet by extracting HOG features, which were then transformed into the color space based on hue, saturation and value for classifying the presence of a helmet. A similar approach of color space transformation was developed by Seong et al. [59] for detecting safety vests, with other classifiers incorporated e.g. support vector machine and logistic regression.

Several algorithms of object detection have been commonly used for detecting workers and PPEs, e.g. the faster region-based convolutional neural network (Faster RCNN) [10]. Using Faster RCNN, Tang et al. [60] detected workers and PPEs separately, where the proximity between the bounding boxes was exploited to determine whether each worker wore a helmet. Fang et al. [61] instead developed an end-to-end method, using the Faster RCNN to directly detect the workers not wearing helmets from

raw video frames. Furthermore, Saudi et al. [17] detected multiple PPEs with Faster RCNN simultaneously, including helmets and vests, for a more comprehensive inspection of safety compliance. In addition, another algorithm named You Only Look Once (YOLO) was adopted by Nath et al. [18] to detect workers along with helmets and vests simultaneously, in which they embedded and compared different CNN variants as the backbone of feature extraction. Overall, CNN-based methods can directly process the image of a worker and classify the status of wearing or missing PPEs, via an end-to-end inference process.

Nevertheless, the existing studies mainly focused on analyzing different model architectures for classifying PPE statuses, while overlooking another problem regarding the dataset structure, i.e. imbalanced samples among classes, which notably degraded the PPE classification accuracy in a recent study by Nath et al. [18]. The majority of their image dataset captured workers wearing PPEs properly, which probably biased the feature learning of their model towards such conditions. Yet, due to insufficient samples among other PPE classes, their model could not learn robust features to identify workers not wearing PPEs, i.e. non-compliance with site safety. For example, quite a few workers not wearing a helmet were overlooked [18], which is undesirable for compliance monitoring. Class imbalance in a dataset is possibly the root cause, which in practice appears inevitable: since wearing PPEs on site is mandated by safety regulations, with-PPE images are much more available than without-PPE ones, i.e. training samples of without-PPE would be limited for developing robust models. However, not wearing the necessary PPEs is an unsafe behavior that is crucial to be identified on sites, which necessitates a methodology to guarantee robust performance on the minority classes despite limited training samples.

3. Methodology

Previous approaches of safety compliance monitoring are limited to individual camera, such that any misbehavior of the workers cannot be continuously tracked over time. Therefore, this study aims to integrate worker ReID and PPE classification into a single framework to enable more comprehensive safety compliance monitoring, as summarized in Fig. 1. After obtaining videos from multiple cameras, Step 1 first detects each worker from the raw images as cropped bounding boxes using computer vision techniques. In Step 2, a ReID model performs feature extraction on each detected worker, from which discriminative human features are exploited for matching the worker identities across multiple cameras. For more robust ReID performance, a new loss function named similarity loss is developed to facilitate the model training process, i.e. learning more discriminative human features. In Step 3, a PPE classification model classifies the PPE statuses from the images of each worker. To improve PPE classification against the problem of imbalanced class samples, a weighted-class strategy is incorporated into the training loss function of a classification model. Finally, in Step 4, the matched identities and PPE statuses of workers are combined for monitoring their safety compliance, where any missing-PPE incidents are logged in a site diary for regular reporting.

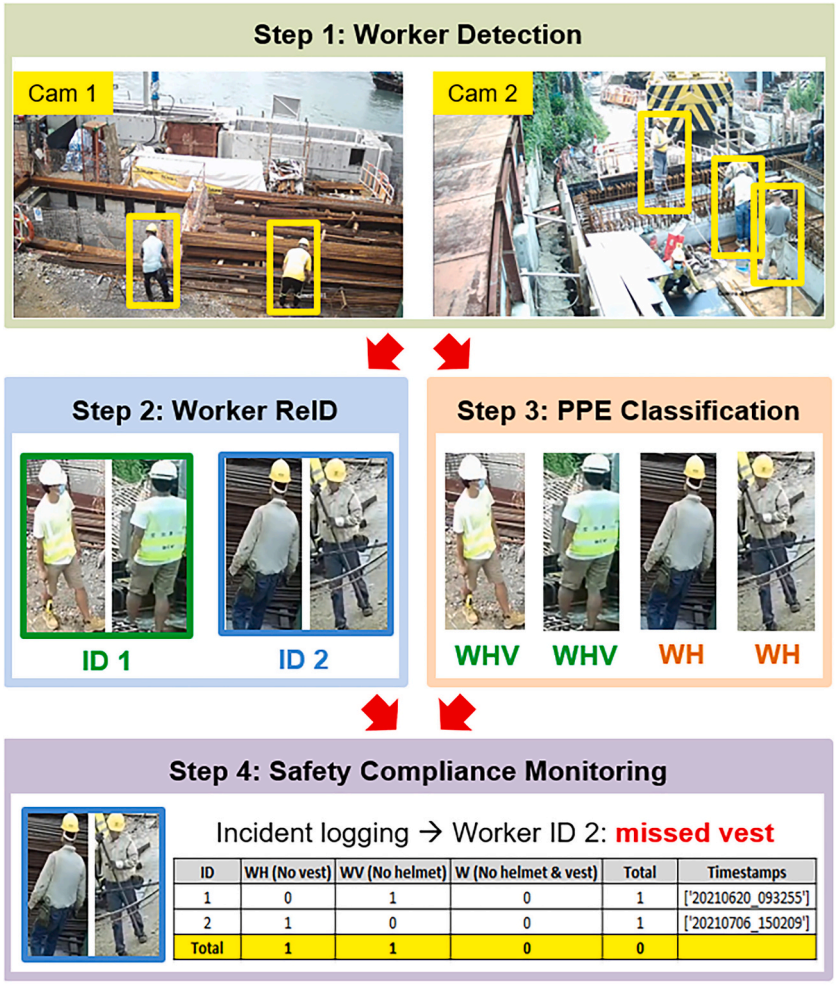


Fig. 1. Overall framework of monitoring the safety compliance of construction workers.

3.1. Worker detection

Upon receiving the raw videos from multiple cameras, the objects of interest are identified in Step 1. With videos collected from a construction site, each construction worker appearing on a frame is detected and cropped as a rectangular bounding box. The detected bounding boxes are further processed by a ReID model (Step 2) and PPE classification model (Step 3). Worker detection is a crucial step that demands an automated process and accurately bounded objects (i.e. cropping the portion of image around a worker with minimal background clutter). There have been numerous studies specialized in worker detection [28,62,63], as well as comprehensive surveys of various state-of-the-art object detection methods [64,65].

Hence, this paper adopts a competitive detection model for the worker detection step. In particular, the YOLO algorithm is used in view of its detection accuracy and inference efficiency revealed in previous studies [18]. It processes a raw image and output a set of bounding boxes that capture each worker. As shown in Fig. 2, each bounding box is associated with four values: (1) timestamp, (2) camera index, (3) unique worker ID, and (4) PPE class index. The first two values are directly obtained from the metadata of the raw video being processed. A timestamp denotes the exact date and time of the video frame captured, while the camera index locates which camera the frame is taken from. The third and fourth values are manually labeled for training the ReID models in Step 2 and the PPE classification models in Step 3 respectively. A worker ID is a unique number assigned by the ReID algorithm that represents the identity of each worker captured. As for PPE status, 'H' and 'V' denote helmet and vest respectively, which is/are appended to the PPE index of a worker if properly worn (e.g. WHV denotes a worker with both helmet and vest). In the testing stage, the worker ID and PPE class index will be inferred by the developed methods. The cropped bounding boxes constitute the input data in the subsequent steps. Since this study focuses on improving the performances of ReID and PPE classification, the term 'image' denotes a bounding box cropped from a raw image hereinafter.

3.2. A new loss function for robust worker ReID

Upon obtaining a set of bounding boxes of the workers, Step 2 performs worker ReID, i.e. extracts identity-discriminative features from each box for matching the same workers across multiple cameras and differentiating different workers. The feature extraction step is particularly essential, from which the feature vectors obtained will be compared, and directly quantify the visual similarities among the workers for identity matching. Many studies have proposed different ReID models, where the recent works mostly designed novel CNN architectures to extract discriminative features. Yet, development of loss functions for facilitating the model training process is relatively underexplored, which is crucial to regulating the learning process of feature representation.

Bounding box		
Timestamp	20200701_093059	20201015_150501
Camera index	2	4
Worker ID	8	9
PPE class	WH	WHV

Fig. 2. Numerical representation of a worker bounding box detected from a raw image.

As a major contribution of this study, a new loss function is developed for a more effective training process of the ReID models. The newly designed method is named 'similarity loss'. Since ReID is performed by comparing feature similarity among different identities, similarity loss directly incorporates the computed similarity scores among different images into the training objective. The proposed loss function is differentiated from several loss functions typically used for ReID model training. Unlike the softmax loss where a classification model is learned from the discrete identity labels, the similarity loss guides a ReID model to learn feature representations that generate consistent similarity scores, i.e. the training objective better aligns with the testing process of ReID. Compared with the hard triplet loss which only considers the relative similarity difference between one positive and negative pair, the similarity loss exploits the similarity scores among all pairs. The following subsections will elaborate the formulation of similarity loss.

3.2.1. Computation of groundtruth similarity labels

For a ReID model to learn consistent similarities among a set of training images, the groundtruth similarity scores of each image pair should be prepared, which are then compared against the one generated by a model, i.e. computing the similarity loss as elaborated in Section 3.2.2. Intuitively, the groundtruth should encourage a model to output high similarity scores among positive pairs (images of the same identity), and low scores among negative pairs (images of different identities). One simplified strategy is to consider a similarity score as a probability value ranging between 0 and 1, i.e. all positive pairs have a similarity score of 1.0 while the negative pairs have a score of 0.0. A model is possibly guided to learn features that can output reasonable similarity scores among positive and negative pairs.

However, such a formulation of groundtruth similarities may force a model to learn a rather aggressive feature representation which maximizes, as much as possible, the relative similarity difference between positive and negative pairs. The potential effect of model training is two-sided. The model may learn discriminative human features that can confidently match the same worker with a similarity score of 1.0 and differentiate different workers with a score of 0.0. Yet, it is also possible that some redundant or even unrelated features are exploited in order to produce the groundtruth similarities, which may cause misleading results of identity matching. For example, if attending to some background clutter like steel bars rather than the worker of interest, different identities may be wrongly matched due to the similar background. Hence, a more robust formulation is needed to address such a deficiency.

The labeling of similarity scores among positive and negative samples is analogous to the labeling process of object classification, i.e. distributing the groundtruth probabilities among positive and negative classes: Among K possible class labels for an image, a straightforward one-hot label gives the positive class a probability of 1.0, while all other negative classes have 0.0. However, making such a large relative difference may lead to over-fitting, since a model is forced to learn features that vigorously increase the confidence of its predicted class label. Certain unrelated features may consequently be learned which do not generalize well beyond the training images. Hence, Szegedy et al. [66] proposed a label smoothing technique as a remedy, which extracts a small portion of probability and uniformly distributes to all negative classes, as illustrated in Fig. 3. For example, 10% of the total probability is extracted from the groundtruth class, such that each of the four remaining negative classes receives a 2.5% probability label. This effectively regularizes the training process, such that a model is prevented from learning misleading features.

Inspired by the label smoothing technique, a smoothing parameter P is incorporated into the formulation of the proposed similarity loss, for a

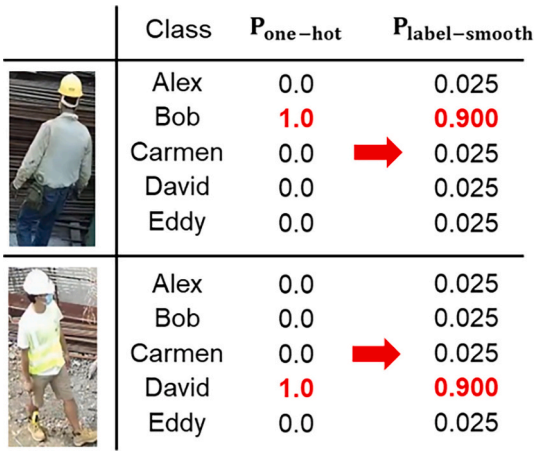


Fig. 3. Illustration of label smoothing technique for object classification.

more flexible definition of similarity labels. Fig. 4 illustrates an example of the similarity labels generated among every pair of training images. Considering the pair-wise similarity matrix of a mini-batch of B images, each of the positive pairs is labeled with a similarity score P (a decimal between 0 and 1), while the negative pairs have a similarity score $1 - P$. Hence, every similarity label has a finite range between 0 and 1, which more intuitively reveals the effect of using different P values in Section 4. Essentially, P can control the relative difference in the groundtruth similarities between positive and negative pairs. For example, if P is reduced to 0.7, the positive and negative pairs have groundtruth similarities of 0.7 and 0.3 respectively, i.e. a much smaller relative difference than labeling with 1.0 and 0.0. This is analogous to the original label smoothing, and possibly different effects can be achieved via tuning P. By defining P as the groundtruth similarity of positive pairs, the

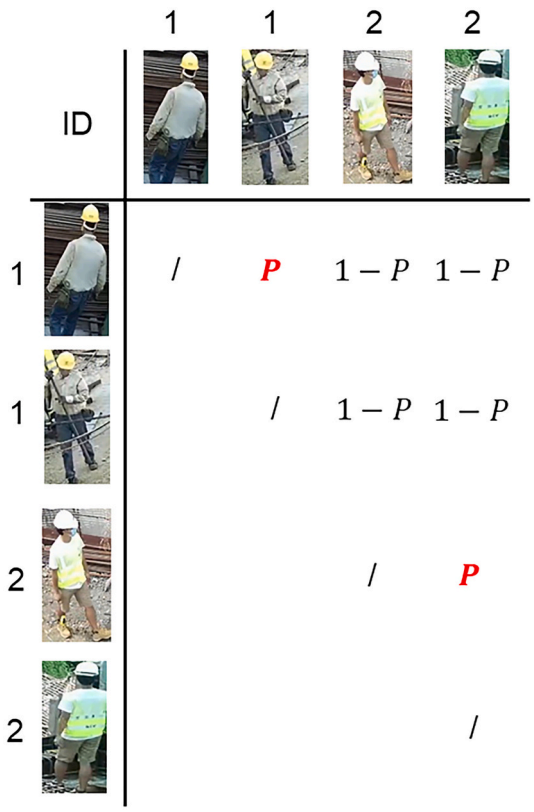


Fig. 4. Illustration of similarity labels generated for a mini-batch of training images.

similarity loss is made generic and adaptable to different ReID scenes upon adjusting the smoothing parameter P.

The generation of similarity labels are automatic, by comparing whether the identity labels of each image pair are the same or different. The pair-wise $B \times B$ similarity matrix then constitutes the groundtruth of the mini-batch during training, guiding a model to exploit features that best replicate the groundtruth similarity scores among the training images.

3.2.2. *Proposed similarity loss for ReID model training*

With the groundtruth similarity labels generated, the training images are fed into the ReID model for feature representation learning. The typical paradigm of supervised learning for a deep CNN is adopted, i.e. the model generates the output (forward propagation) whose deviation from the groundtruth is computed as the training loss to iteratively update the model (backward propagation). Considering a typical CNN architecture for ReID task, a feature extraction backbone comprises several convolutional, pooling and fully connected layers, for exploiting hidden features from the images. Given a mini-batch of B images, each image is represented by a feature vector extracted by the backbone model, whose feature dimension depends on the model architecture.

Subsequently, the visual similarities among the B images are compared via correlating the B feature vectors extracted. More specifically, a $B \times B$ pair-wise similarity matrix is computed among the feature vectors, as exemplified in Fig. 5. Generally, different metrics are applicable to computing the similarity scores. In this study, cosine similarity is adopted which benefits the computation of similarity loss by enforcing a finite range between 0 and 1 to the similarity scores, aligning with the defined range of the similarity labels. Each element (i,j) in the similarity matrix is computed according to Eq. (1), i.e. the cosine similarity between feature vectors i and j. Since all elements along the diagonal axis denote the self-similarity of a vector itself (which are always 1.0 due to the equivalence), they are ignored when computing the similarity loss.

$$cosine\ similarity\ (i,j) = \frac{f_i \cdot f_j}{\|f_i\| \times \|f_j\|}\ if\ i \neq j, else\ Null \tag{1}$$

Based on the metric above, two similarity matrices are compared during training with the similarity loss: (1) M computed via the forward propagation of the ReID model, and (2) M_{GT} generated as the groundtruth. To perform backward propagation for model updating, the similarity loss is computed according to Eq. (2). Mathematically, the element-wise squared differences between M and M_{GT} are first computed. A matrix with unchanged dimension is then obtained, which captures how much the similarity scores of each image pair deviate from its groundtruth. Subsequently, the largest value is extracted by the *max*() operation, i.e. the maximum deviation of the similarity score is defined as the similarity loss value among a mini-batch of training images. Intuitively, this loss function aims to force the feature representation of a model to gradually reconstruct the groundtruth similarity scores, i.e. higher similarity among positive pairs, and lower among negative ones. Compared with the softmax loss which formulates ReID training as a classification task, the training objective of the similarity loss better aligns with the testing process of ReID, such that the model receives more direct supervision during its training.

$$similarity\ loss = max\left((M - M_{GT})^2\right) \tag{2}$$

Regarding the loss function design, the differences are squared such that the absolute magnitudes are considered. Several variants have been tested, e.g. absolute differences, among which the squared differences yielded the best performance. Moreover, the *max*() operation effectively suppresses the deviation of the model output from the groundtruth. Other variants have also been tested, e.g. *mean*() which takes the averaged deviation among all image pairs in a mini-batch. The *max*() showed the most optimal performance among the attempted variants. Therefore, the strategy of taking the maximum squared difference constitutes the

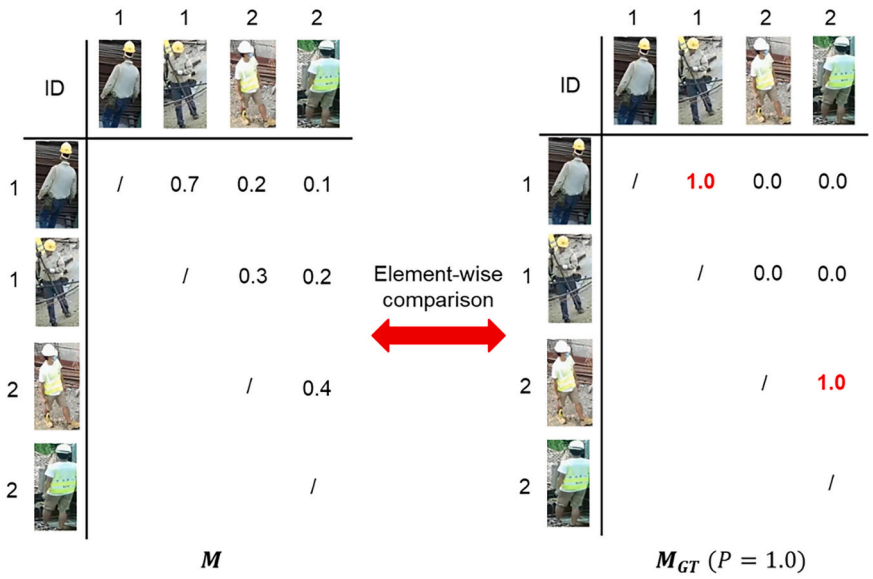


Fig. 5. Illustration of $B \times B$ similarity matrix among a mini-batch of B images.

proposed similarity loss for ReID model training.

3.2.3. Identity matching with ReID features

In the testing stage, given the feature vectors extracted by the developed ReID model, a worker identity should be associated with each image which is kept consistent across different cameras. More specifically, a methodology is needed to accurately give a unique identity to all the images of the same worker, and simultaneously to differentiate different workers. This task of identity matching is formulated as a clustering problem based on the extracted feature vectors. To this end, the methodology recently proposed by Wong et al. [67] is adopted, i.e. an improved version of the hierarchical agglomerative clustering algorithm [68]. Referring to previous ReID studies, the identity matching process greatly relies on the quality of the features extracted [69], i.e. how identity-discriminative each feature vector is. Hence, this study focuses on developing a more robust ReID method, i.e. the newly designed similarity loss. A baseline method is adopted for the identity matching, such that the improvement brought by the proposed ReID loss function is directly revealed.

3.3. A new strategy for robust PPE classification

In addition to the unique identities, the PPE status of each worker is identified. Based on previous studies, necessary PPEs including helmet and vest are needed for working on site most of the time. Hence, classifying whether workers are carrying these two PPEs will enable the monitoring of routine compliance to site safety.

3.3.1. Baseline methodology

Identification of the PPE status of a worker is formulated as a multi-class classification problem, i.e. each worker is classified as one of the pre-defined PPE statuses. To this end, the methodology by Nath et al. [18] is referenced. They proposed several vision-based approaches for detecting helmet and vest worn by a worker, where Fig. 6 illustrates the baseline workflow adopted in this study for PPE classification. Specifically, given a bounding box that captures a worker, a CNN is deployed as a classifier to process the cropped image and output the PPE status of the worker. To this end, four PPE classes are defined: (1) W – worker not wearing the two necessary PPEs, (2) WH – worker with helmet only, (3) WV – worker with vest only, and (4) WHV – worker with both helmet and vest. Consequently, PPE classification is formulated as a classification problem, with cropped worker images as the input.

Nevertheless, baseline studies such as Nath et al. [18] did not address the problem of imbalanced samples among classes, which notably degraded the PPE classification accuracy in the experimental dataset [18]. Among the four classes in their dataset, 'WH' and WHV' constitute the majority of the image samples (i.e. more than 90%). This probably biased the feature learning of a model towards those majority classes, since the model could not exploit features from images of other classes due to insufficient samples. Consequently, all the trained CNNs showed much lower classification accuracies particularly for 'W' and 'WV', which were the minority classes but were misclassified. For example, more than 30% images of 'WV' were classified as 'WHV'. In other words, quite a few workers without helmets were undesirably overlooked, rendering the process of compliance monitoring unreliable. Since class imbalance in the dataset is possibly the root cause, a methodology is needed for robust performance on the minority classes under limited training samples.



Fig. 6. Baseline workflow of PPE classification (modified from [18]).

3.3.2. Proposed weighted-class strategy against imbalanced samples

On top of the baseline workflow, a weighted-class strategy is developed to facilitate the model learning towards minority classes. Specifically, the weights of each class are strategically manipulated when their samples participate in the backward propagation of a model, i.e. control how much each sample influences the model updating. The general principle is to increase the weights of the samples in the minority classes, such that they individually impose larger influence than the ones in a majority class. Apparently, quantifying how much to increase the weight is essential. In particular, the proposed strategy of class weighting is shown in Eq. (3), which considers the ratios of the sample counts among all classes. Formally, given the number of samples among each class (S_c), the largest number is first found by $\max_c S_c$. Then, within each class c, a new class weight is assigned by dividing $\max_c S_c$ by the sample count of class c (S_c). The new weight W_c is then assigned to every sample of class c during model training. Consequently, the samples will enforce different extents of contribution to the loss function in each mini-batch. Since an ordinary classification problem is formulated, the softmax loss is adopted where each sample is weighted according to its new weight.

$$\text{new class weight}, W_c = \frac{\max_c S_c}{S_c} \tag{3}$$

The proposed strategy of computing class weights aims to equalize the aggregated contribution among all classes, given a dataset with imbalanced numbers of samples. Numerically, the new weight of the class with the largest sample count is 1.0 ($\max_c S_c = S_c$), while that of other classes with fewer samples are increased proportionally to the ratios of the sample counts. For example, a half-sized class will receive a new weight of 2.0, such that each sample has a doubled influence on a model relative to that in the largest class. Generally, for every class c, the product $S_c \times W_c$ yields the same value of $\max_c S_c$. This intuitively depicts that the total weighted influences over all the samples are equivalent among each class. Given imbalanced samples among different classes, the problem of biased training is mitigated by balancing the weighted influences. Hence, the proposed weighted-class strategy is incorporated into the model training for more robust PPE classification, especially for the minority classes with limited training samples.

3.4. A workflow for integrated safety compliance monitoring

With the developed methodologies of worker ReID (Step 2) and PPE classification (Step 3), the unique identity and PPE status are respectively obtained for each worker image. In Step 4, these two sources of results are combined to monitor the PPE compliance of each worker over time. Any incident of missing the PPEs is logged to generate regular reports, based on which site officers can review the on-site behaviors of workers and devise remedial measures for enhanced safety awareness.

With the unique identities labeled, all the worker images can be clustered such that each cluster includes the images belonging to the same worker. Since each image itself contains the timestamp, the trajectory of each worker can be continuously tracked over time. On top of that, the frame-by-frame PPE status is considered, such that the PPE status of a worker is monitored over time (i.e. how many frames each status has elapsed). Eventually, any incident of PPE non-compliance can be identified from the results above. To enable a more practical monitoring process, an incident counting mechanism is designed whose pseudo code is summarized in Fig. 7. After processing the images, a set of bounding boxes, each labeled with a timestamp, worker identity and PPE status, are obtained. Incident counting is then performed for each individual worker, i.e. the boxes belonging to each worker are first extracted and sorted in a chronological time series. For each worker, a state tracker and incident counter are initialized. By processing the labeled information of each bounding box, the state tracker and incident counter are updated upon meeting certain criteria defined below for recognizing an incident.

For deciding how an incident is counted, two parameters are introduced namely (1) minimum buffer time T_b and (2) maximum reset time T_r. To exemplify the procedure, Fig. 8 illustrates different cases of incident counting governed by these two parameters. First, T_b imposes a minimum period as the buffer time before starting to count one incident. It aims to make the incident logging more reasonable by inserting a mechanism of delayed counting, since a worker may for example take off the helmet for a while to adjust its tightness, under safe surrounding

```
Input:
    1. All processed bounding boxes each labeled timestamp t, identity i, PPE status p
    2. Time thresholds T_b, T_r

For each identity i:
    Boxes = {bounding boxes labeled i, sorted by timestamps}
    t_b, t_curr, count = 0
    state = negative
    For each box in Boxes:
        If t_curr == 0:
            t_curr = t   #Initialize tracker of current t at the first frame
        If p ∈ {'W', 'WV', 'WH'}:
            If state == negative:   #Check minimum buffer time
                t_b = t_b + (t − t_curr)
                If t_b ≥ T_b:
                    state = positive
                    count = count + 1
                    t_b = 0
            Else:   #Check maximum reset time
                If t − t_curr ≥ T_r:
                    state = negative
        Else:   #Reset state if worn the necessary PPEs
            state = negative
            t_b = 0
        t_curr = t
```

Fig. 7. Pseudo code for the mechanism of incident counting.

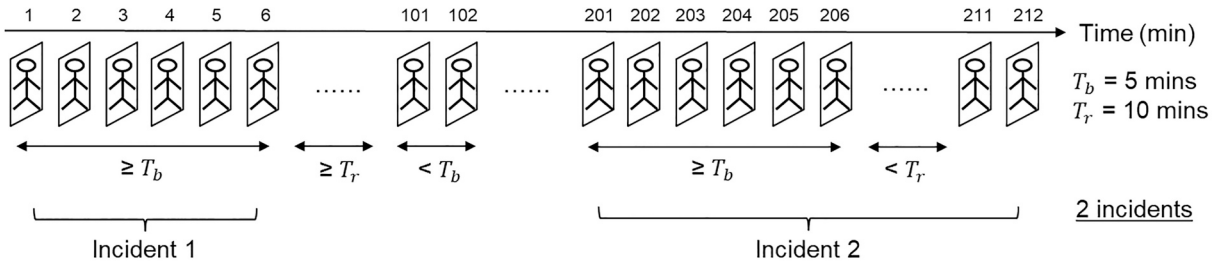


Fig. 8. Illustration of two parameters (T_b and T_r) for incident counting.

conditions. Without a buffer time, penalizing a worker not wearing the PPEs in every frame may be overly strict for compliance monitoring. Formally, based on the PPE status identified in each timestamp, a worker is counted towards an incident only after the 'without PPE' status is maintained for T_b timestamps. In the pseudo code, a timer t_b is set to track the duration of a worker not wearing any of the necessary PPEs.

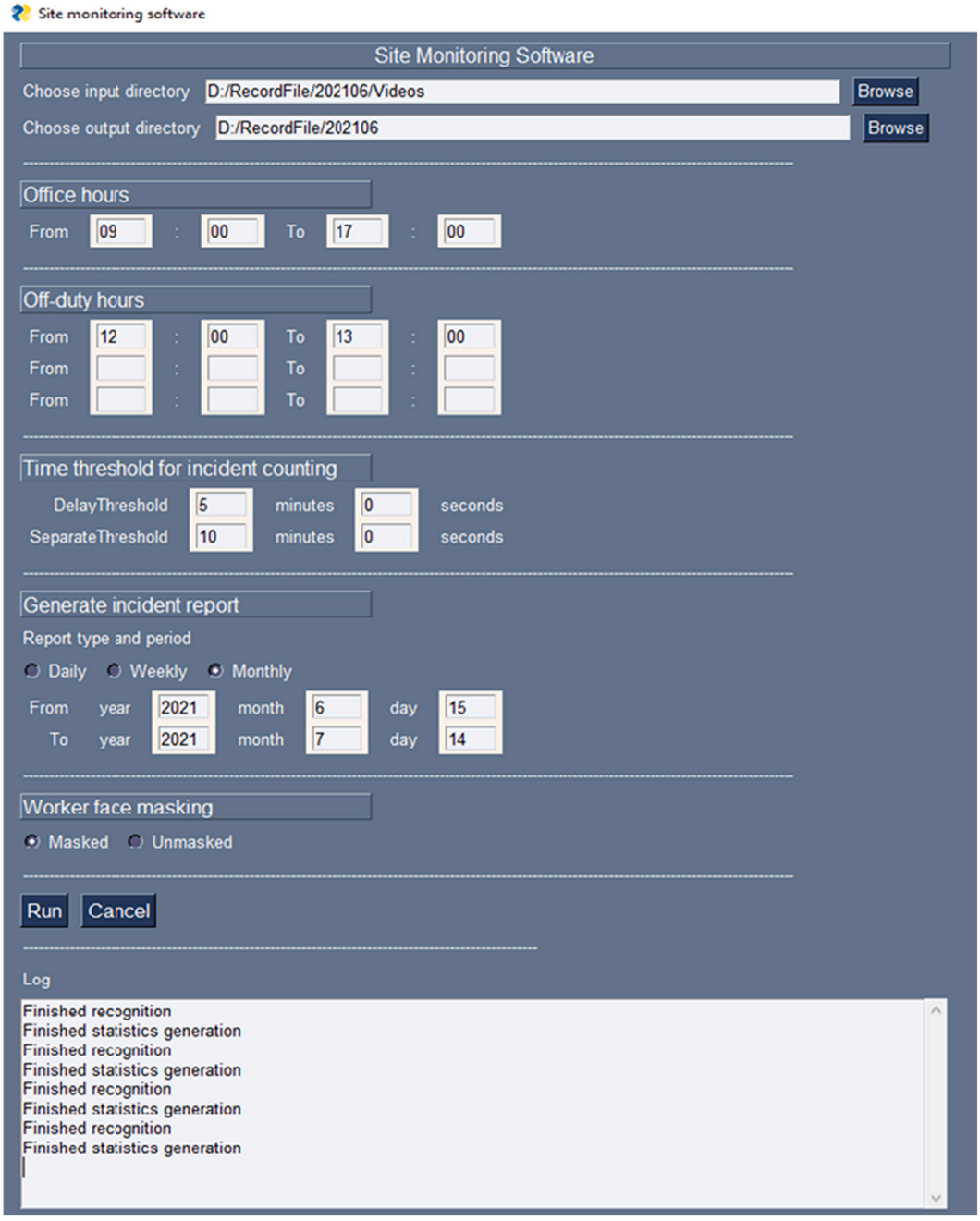


Fig. 9. User interface of the developed software for automated site analytics.

When attaining the minimum buffer time (i.e. $t_b \geq T_b$), the state of a worker becomes 'positive' and the incident count is incremented. Otherwise, the state returns to 'negative' until the next frame that captures a without-PPE status.

In addition, another parameter T_r is set to separately count two incidents with a long time gap in between. The maximum time gap defines a reasonable duration beyond which an incident terminates, i.e. any 'positive' state is reset if not observed for a certain time. For example, a worker not detected for T_r timestamps may have left the scene such that the PPE status is unknown within the time gap (e.g. back to the site office with no PPE required). By the time the worker reappears, any missing-PPE behavior should be unrelated to the previous incidents and hence counted as a new one. Formally, if two boxes with 'positive' status are separated by the pre-defined time gap (i.e. $t - t_{curr} \geq T_r$), the counter of the current incident is reset, and any positive status in future timestamps will possibly be counted as another incident upon meeting the minimum buffer criteria.

To facilitate site monitoring in practice, a user-friendly software is developed to deploy the ReID and PPE classification methods in a single toolkit. Fig. 9 illustrates the user interface of the developed software, showing the overall workflow and functionalities for site analytics in practice. The software can be run on commercial computers preferably with accelerated computing resources such as graphical processing units.

To start with, users first choose the data folder which stores the raw input videos, and another folder for saving the output data generated by the software. Then, the on-duty hours are pre-defined for selectively processing the videos within certain periods. For example, videos in normal working hours 9:00–17:00 are processed, while that within lunch hour 12:00–13:00 may be ignored when site operations are mostly suspended, resulting in minimal safety hazard. Furthermore, users can define the time span and the type of incident report to be generated. For example, by picking a 1-month period, the incidents identified within this period are reported, where the incident statistics may be aggregated daily, weekly or monthly chosen by users. For visual inspection by safety officers, video frames capturing the incidents are extracted to a local database. For instance, safety officers can authenticate whether the reported incidents are reasonable, by inspecting the actual behavior of workers on the frames. This manual review process ensures that the compliance monitoring is reliable and a justified incident report is obtained eventually. To preserve the privacy of workers, their faces can be automatically blurred upon saving the video frames. By clicking the 'Run' in the program interface, video processing starts and the results are saved automatically.

Overall, accurate incident counting is based on the robustness of ReID and PPE classification, to which the methods proposed in Sections 3.2 and 3.3 can contribute. Along with the developed procedure in this sub-section, the labeled identities and PPE statuses of the worker images are summarized into site incident reports for the safety officers, e.g. to evaluate the safety compliance of workers and devise any remedial measures needed.

4. Experiment design and results

4.1. Experimental design

The proposed computational methods and the overall monitoring system were implemented in a commercial computer, with one NVIDIA GTX1080 graphical processing unit (GPU) and a 4-core i7–6700 CPU at 3.40GHz.

4.1.1. Baseline models

Since this paper focuses on developing new loss functions for model training, several typically used CNNs are selected as the baselines. For the ReID part, three models are studied, namely ResNet50 [11], OSNet [47] and OSNet+BDB [67]. ResNet50 is typically used as the backbone network for various visual recognition tasks, while OSNet and OSNet+BDB are two ReID-specific models that recently achieved competitive performances in several public benchmarks. They are chosen mainly because their learned feature representations for the ReID task are notably different. As revealed by Wong et al. [67], these three models exhibit very different mechanisms of spatial attention, i.e. they focus on different regions of a person when extracting ReID features. Hence, by integrating the similarity loss into them, the generalizability of the proposed loss function to different network architectures can be validated. As for the PPE classification, the proposed weighted-class strategy is deployed into two CNN backbones, namely VGG16 [39] and ResNet50 [11]. They were used in a recent study [18] as the baselines of PPE classification, while the class imbalance problem was not addressed. Hence, studying these two CNNs can effectively validate the improvement by deploying the proposed weighted-class strategy.

The hyper-parameters for model training are configured, as shown in Table 1, which mostly refer to previous studies of the baseline models [18,67]. For ReID, each model is trained for 50 epochs and the model weights are saved every 5 epochs, after which the model achieving the best validation accuracy is kept for subsequent evaluation. Different combinations of loss functions are tested, where the contribution of adding the proposed similarity loss is demonstrated in Section 4.2. As for the PPE classification, the training setup basically follows [18]. As a typical classification task, the softmax loss is adopted for training PPE classification models, with and without the proposed weighted-class strategy incorporated.

4.1.2. Datasets

An in-house dataset is created from a real construction site in Hong Kong, for validating the developed methods for ReID and PPE classification. The construction project was led by the Drainage Service Department (DSD) of the Hong Kong government, which involved building drainage facilities near a waste water discharge area. As shown in the layout plan in Fig. 10, four surveillance cameras were installed at heights around 15 m. The cameras captured the major working and walking areas on the site with non-overlapped fields-of-view. Some sample images of the workers are shown in Fig. 10, which illustrate several ReID challenges, e.g. large variations of the image scales and worker poses. Since the workers were mostly wearing helmet and vest, different workers have a fairly similar visual appearance. These constitute intra-person variation and inter-person similarity, making this DSD dataset sufficiently challenging to validate the robustness of the developed ReID method.

Video recordings over one week were obtained and sampled for the labeling of worker identities and PPE statuses. Totally 6245 cropped worker images belonging to 40 unique workers were labeled. The dataset was then split into training and testing sets, as summarized in Table 2. For worker ReID, 3735 images among 24 workers were used for training the ReID models, and tested with 2510 images among the other 16 workers. As for PPE classification, each worker image was classified into four PPE statuses, where a significant imbalance of samples was observed among the different classes. The majority classes were 'WH' and 'WHV' accounting for more than 90% of the images, among the minority classes 'W' and 'WV'. Such a sample imbalance can validate the effectiveness of the developed weighted-class training strategy.

Table 1
Hyper-parameters for model training for ReID and PPE classification respectively

Hyper-parameter	ReID	PPE classification
Image size (H × W)	256 × 128	256 × 128
Batch size	32	32
Epochs	50	25
Optimizer	AMSGrad	Adam
Loss functions	{Softmax, Hard triplet, Similarity}	{Softmax}
Base learning rate	0.00015	0.0001

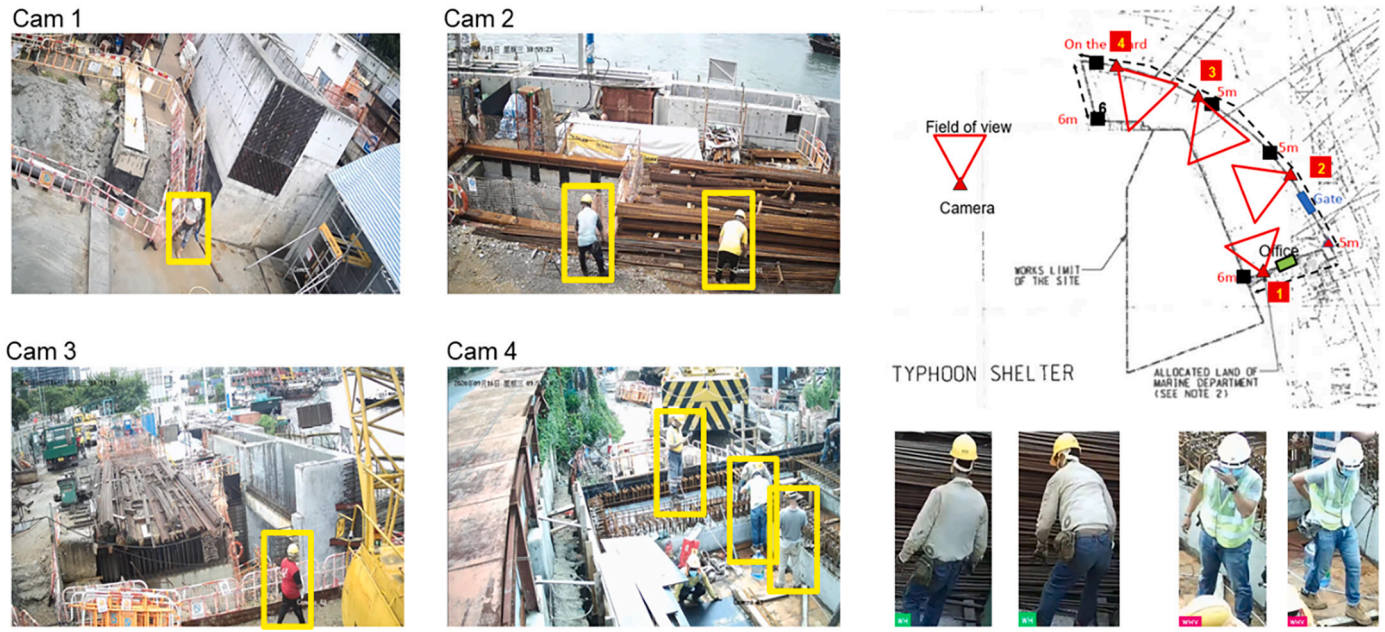


Fig. 10. Camera layout and sample images in different cameras on the DSD site.

Table 2
Statistics of the in-house construction site dataset for ReID and PPE classification

PPE classes	No. of images	
	Train	Test
W	30	20
WH	1190	780
WV	100	60
WHV	2415	1650
Total	3735 (24 workers)	2510 (16 workers)

To evaluate the generalizability of the proposed ReID method to different scenes, the results in a public ReID benchmark were also obtained, namely ReID Across Indoor-outdoor Dataset (RAID) [70]. Its images were collected among four cameras, with two indoor and two outdoor capturing urban streets. The dataset scale of RAID is comparable with ours which was collected on a construction site. As summarized in Table 3, the two datasets have fairly similar numbers of images and unique identities, while RAID captures several generic urban scenes, among which the human appearance is more diverse than that on a construction site, especially in regard to clothing. As shown in Fig. 11, significant variations of clothing styles, lighting and human poses are observed among different persons, which require the ReID models to extract robust features. By evaluating the ReID method with these two datasets, its generalizability in different scenarios can be validated.

4.1.3. Evaluation metrics

By undertaking person ReID among a group of images, unique identities are mapped to each image. Therefore, the identity mapping accuracy of a model is evaluated in Section 4.2 by the identification F1 (IDF1) score, which is a common metric for human tracking across multiple cameras [71]. The mathematical representation of the IDF1 score is summarized in Eq. (4), which computes the harmonic mean of the precision and recall of the identity mapping results. The IDF1 score measures how consistently the identity labels are assigned to the images, i.e. a higher IDF1 score reflects that the images of the same person receive a consistent identity, while that of different persons receive distinct identities.

$$IDF1 = \frac{2*IDTP}{2*IDTP + IDFP + IDFN} \tag{4}$$

$$F1_c = \frac{2*TP_c}{2*TP_c + FP_c + FN_c} \tag{5}$$

$$F1 = \frac{1}{C}\sum_{c}^{C} F1_c \tag{6}$$

As for the PPE statuses, the F1 score is used for evaluating the classification accuracy in Section 4.3. First, given C pre-defined classes, the F1 score for each class is computed to report class-wise performance, as shown in Eq. (5). Subsequently, a macro-averaged F1 score is computed by averaging the class-wise F1 scores, as in Eq. (6), resulting in an aggregated indicator of the classification accuracy of a model. Since macro-averaging is independent of the sample sizes among each class, the majority classes would not dominate the average F1 score, i.e. macro-averaging can emphasize the performance of the minority classes [72]. Therefore, this metric can validate the robustness of the PPE classification method against the class imbalance problem.

These two metrics can reflect the overall reliability of the compliance monitoring system, i.e. accuracy of reporting any incident. For worker ReID, a high IDF1 score depicts that the worker identities are accurately assigned. Consistently matched identities are essential, since any misbehavior of the same worker should be tracked over time, of which the duration will be compared against certain time thresholds. If the identities are switched among different workers, a wrong duration of misbehavior will be computed, such that false positives/negatives of incident may be reported eventually. Similarly, the F1 score for evaluating PPE classification serves to indicate how accurately the PPE status is predicted in each frame. If any missing-PPE status is overlooked, the estimated duration of misbehavior will also be inaccurate, making it unreliable for the duration-based incident reporting. In summary, robust performances in worker ReID and PPE classification are significant to

Table 3
Statistics of the two ReID datasets used

Dataset	No. of images	
	Train	Test
Site (ours)	3735 (24 IDs)	2510 (16 IDs)
RAID	3635 (22 IDs)	3285 (21 IDs)



Fig. 11. Sample images of the ReID across indoor-outdoor dataset.

the developed compliance monitoring system.

To evaluate the overall applicability of the proposed mechanism, the False Discovery Rates (FDRs) for incident reporting under different time thresholds are also reported in Section 4.4, upon processing a video with the proposed monitoring system end-to-end. For such study, FDR is defined in Eq. (7) according to [73], i.e. the proportion of false positives (FP) among the sum of false and true positives (TP), where "positive" is defined as a reported incident in this study. To compute the FDR given a minimum buffer time T_b, the number of reported incidents and correlated frames are first extracted after the system processes a video. These results are then compared against the actual PPE status on the frames. For each no-PPE incident reported, if the worker on the triggering frame actually wears the necessary PPEs, such incident is marked as falsely reported. Eventually, the ratio of the falsely reported incidents (FP) to all the reported ones (FP + TP) is computed as FDR. If no incident is reported, FDR equals zero [73]. Intuitively, as T_b increases, the chance of reporting an incident decreases because a longer buffer time is granted to each worker. Hence, evaluating FDR under different T_b will reveal the sensitivity of such time threshold.

$$FDR_{T_b} = \frac{FP}{FP + TP} = \frac{No.of\ incidents\ \textbf{falsely}\ reported}{No.of\ incidents\ reported}, under\ buffer\ time\ T_b \tag{7}$$

4.2. Results of the proposed similarity loss for ReID

This section reports the evaluation results of the similarity loss, showing its (1) improvement over several existing loss functions used for training different ReID models, (2) generalizability to different ReID datasets, and (3) sensitivity analyses of the smoothing parameter.

4.2.1. Improvement over existing loss functions

Overall, the proposed similarity loss function brings consistent improvement when added on top of several existing losses, e.g. softmax and hard triplet losses. More crucially, such improvement is also generalizable to different ReID models, validating the effectiveness of the similarity loss. The detailed results of using different combinations of losses are summarized in Tables 4, 5 and 6, with different ReID models studied respectively.

Table 4
Improved IDF1 scores by adding similarity loss into ResNet50 in DSD dataset

Variant	Softmax loss	Hard triplet loss	Similarity loss	IDF1 (%)
1			√	95.20
2	√			96.61
2*	√		√	97.89
3		√		96.07
3*		√	√	97.91
4	√	√		97.93
4*	√	√	√	**98.17**

Table 5
Improved IDF1 scores by adding similarity loss into OSNet in DSD dataset

Variant	Softmax loss	Hard triplet loss	Similarity loss	IDF1 (%)
1			√	89.99
2	√			90.39
2*	√		√	93.11
3		√		92.56
3*		√	√	**96.62**
4	√	√		92.67
4*	√	√	√	95.25

Table 6
Improved IDF1 scores by adding similarity loss into OSNet+BDB in DSD dataset

Variant	Softmax loss	Hard triplet loss	Similarity loss	IDF1 (%)
1			√	92.81
2	√			85.40
2*	√		√	87.74
3		√		91.15
3*		√	√	**95.82**
4	√	√		87.99
4*	√	√	√	90.44

For each model and on top of each loss combination, the similarity loss is added to show its contribution. Among all the three models, adding the similarity loss consistently boosts the IDF1 accuracy over the setup without it. For example, OSNet+BDB notably improves from 91.15% to 95.82% after adding the similarity loss on top of the hard triplet loss. There is around 2% improvement for other loss combinations. As for the OSNet trained with softmax loss + hard triplet loss, adding the similarity loss also boosts the IDF1 accuracy from 92.56% to 96.62%. Other loss combinations show 2–3% improvement after adding the similarity loss. As for ResNet50, its IDF1 scores almost dominate the other two models with different loss combinations, and its improvement by adding the similarity loss is relatively less, where the biggest one is from 96.07% to 97.91% with the similarity and softmax losses. Overall, adding similarity loss can effectively improve the ReID accuracy for different models.

Overall, using any of the losses alone only achieves sub-optimal results. The highest IDF1 accuracies of all three models are associated with combinations of multiple losses, e.g. combining the three losses for ResNet50 and OSNet, and using hard triplet + similarity losses for OSNet+BDB. These show the benefit of multi-loss training, since each loss function imposes particular effects in facilitating the feature learning of a ReID model.

4.2.2. Generalizability to different ReID datasets

In addition to the generalizability to different ReID models, the same

evaluation procedure is conducted with another public ReID dataset named RAID, to validate that the proposed similarity loss is generalizable to different scenarios. Tables 7, 8 and 9 summarize the improvements to three ReID models upon adding the similarity loss.

Among all the loss combinations used to train the three models, adding the similarity loss consistently boosts the IDF1 accuracies in the RAID dataset. For example, ResNet50 improves from 82.46% to 86.92% upon adding the similarity loss on top of the hard triplet loss. Compared with the softmax + hard triplet losses, further adding the similarity loss also improves OSNet (from 85.75% to 87.94%) and OSNet+BDB (from 87.91% to 90.91%). Overall, the similarity loss can bring a 2–4% improvement in IDF1 accuracy among the three models in the RAID dataset.

Crucially, the consistently improved performances in both the DSD and RAID datasets validate that the proposed similarity loss is generalizable to different scenarios, including worker ReID on construction sites and pedestrian ReID in indoor or outdoor urban areas.

4.2.3. Parameter sensitivity analysis

As introduced in Section 3, a smoothing parameter P is defined to govern the relative similarities labeled among the positive and negative sample images. This sub-section analyzes the sensitivities of IDF1 accuracy against different values of P, in order to interpret the effect on ReID feature learning upon tuning this parameter. Figs. 12 and 13 illustrate the IDF1 scores upon varying P for the three ReID models, respectively, in the DSD and RAID datasets.

Overall, the ranges of optimal P values differ among different models and datasets. With the DSD dataset, OSNet and OSNet+BDB are fairly sensitive to the varying of P, with the maximum difference in IDF1 scores larger than 6%, while that of ResNet50 is around 2%, particularly showing its lower sensitivity to P in the DSD dataset particularly. Moreover, the three models attain their optimal IDF1 scores at different P values, i.e. 0.9 for ResNet50, 0.8 for OSNet and 1.0 for OSNet+BDB. As for the RAID dataset, ResNet50 and OSNet have relatively smaller variations of IDF1 scores upon varying P, i.e. a 2% difference, while that of OSNet+BDB is more notable with 4% maximum difference. Similar to the DSD dataset, these three models also have different optimal P values in the RAID dataset, i.e. 0.8 for ResNet50, 0.6 for OSNet and 0.7 for OSNet+BDB.

The analyses illustrate that introducing a smoothing parameter P makes the proposed method more generalizable. The proposed similarity loss can be adaptively deployed to different ReID models and datasets, by optimizing P for different network architectures and scenarios. The sensitivity analyses reveal some possible correlations between the P value and these two factors. For network architecture, a larger dimension of the feature descriptor may favor a larger P value. The feature dimensions of ResNet50 and OSNet are 2048 and 512 respectively, and with these values, ResNet50 consistently favors larger P values than OSNet does ($0.9 > 0.8$ in DSD, $0.8 > 0.6$ in RAID). To interpret such a finding, a larger feature dimension tends to narrow down the distances between feature descriptors [68], hence a larger P value possibly forces a model (e.g. ResNet50) to learn more discriminative features that enlarge the feature distances among different persons. To impose such an effect on models with smaller feature dimension (e.g. OSNet), a smaller P value may already suffice, while an unnecessarily large P value may instead lead to overfitting upon feature learning. Overall, the choice of the optimal P value should account for the distribution of the feature distances of a model.

As for deploying in different scenes, a larger P value may be preferred for lower diversity in human appearance (i.e. the features such as clothing among different persons are rather similar). Among the three

Table 7
Improved IDF1 scores by adding similarity loss into ResNet50 in RAID dataset

Variant	Softmax loss	Hard triplet loss	Similarity loss	IDF1 (%)
1			√	76.20
2	√			76.98
2*	√		√	78.84
3		√		82.46
3*		√	√	**86.92**
4	√	√		83.45
4*	√	√	√	84.83

Table 8
Improved IDF1 scores by adding similarity loss into OSNet in RAID dataset

Variant	Softmax loss	Hard triplet loss	Similarity loss	IDF1 (%)
1			√	83.88
2	√			86.14
2*	√		√	87.29
3		√		86.51
3*		√	√	**87.98**
4	√	√		85.75
4*	√	√	√	87.94

Table 9
Improved IDF1 scores by adding similarity loss into OSNet+BDB in RAID dataset

Variant	Softmax loss	Hard triplet loss	Similarity loss	IDF1 (%)
1			√	80.69
2	√			83.53
2*	√		√	85.73
3		√		87.87
3*		√	√	88.33
4	√	√		87.91
4*	√	√	√	**90.91**

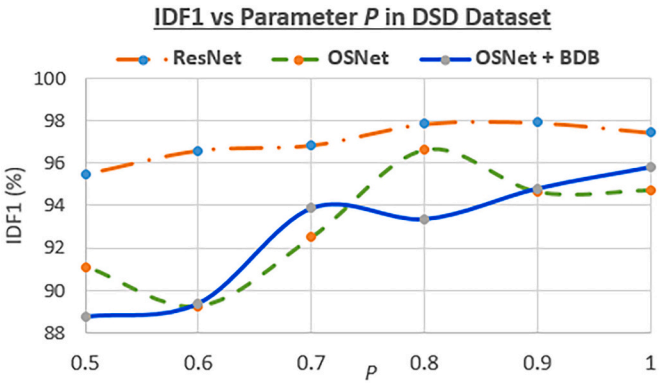


Fig. 12. Sensitivities of IDF1 scores against smoothing parameter P in DSD dataset.

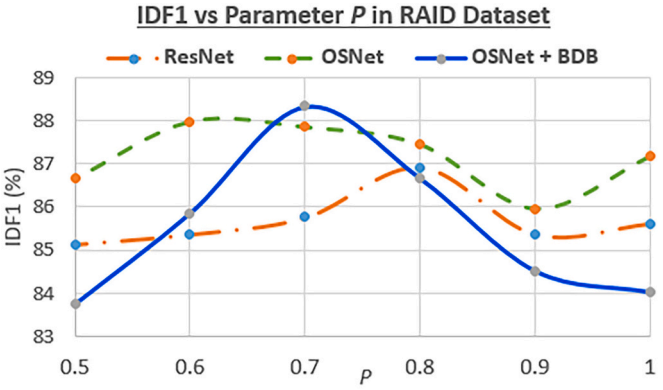


Fig. 13. Sensitivities of IDF1 scores against smoothing parameter P in RAID dataset.

models, the ranges of the optimal P values differ across two different datasets (0.8– 1.0 for DSD, 0.6– 0.8 for RAID). The DSD scenario favors a larger P, possibly because construction workers tend to have very similar clothing, e.g. wearing safety vests and helmets, rendering the feature distances between different persons fairly small. In this case, similar to the feature dimension effect, a larger P can encourage the learning of more discriminative human features. On the contrary, the RAID dataset captures ordinary pedestrians whose clothing styles are more diverse, making the feature distribution naturally more separated. A slightly smaller P value may therefore suffice to effectively facilitate feature learning. Overall, the P value should be optimized against the human appearance in a particular scenario.

4.3. Results of the proposed weighted-class strategy for PPE classification

This sub-section evaluates the performance of PPE classification in the DSD dataset, particularly with the proposed weighted-class strategy against the problem of imbalanced training samples among different PPE classes. To this end, different commonly used CNNs are trained with the proposed class weights imposed, and the results are compared against using equivalent class weights (i.e. not using the proposed loss function). Table 10 summarizes the F1 scores of two CNNs with respect to different PPE classes, with and without the proposed weighted-class strategy.

Overall, the weighted-class strategy notably improves the PPE classification performance, especially for those minority classes (i.e. PPE classes with fairly few training samples). Among both VGG16 and ResNet50, adding the weighted-class strategy boosts the average F1 scores by 5–13% (e.g. VGG16 improves from 74% to 87%). Furthermore, the proposed strategy mostly improves the individual F1 scores towards each PPE class, especially the two minority classes, namely 'W' and 'WV'. The two CNNs show around 12–23% increases in the F1 scores for these two PPE classes. For the 'W' class, VGG16 improves from 50% to 72% while ResNet50 improves from 47% to 59%. For the 'WV' class, the F1 score is improved respectively from 66% to 89% and from 55% to 67%. As for the majority classes, VGG16 also boosts the F1 scores from 86% to 91% for 'WH' and from 93% to 96% for 'WHV'. In summary, the proposed weighted-class strategy notably enhances the PPE classification performance, particularly for the minority classes with limited training samples.

By assigning higher weights to the minority classes, their training samples can impose a larger influence on the learning process, i.e. the batch gradient backpropagation of a model. Despite the small ratios of the training samples, the model is forced to learn features that are generalizable to those minority classes as well as the majority ones. This possibly explains why the F1 scores of the minority classes show more significant improvement. Hence, the proposed strategy adds practical value to worker PPE classification because, as elaborated in Section 2, a minority class probably captures some abnormal or unsafe actions of workers that are rarely observed on-site, e.g. not wearing PPEs, which are particularly important to be identified. Therefore, the improved PPE classification on the minority classes can contribute to more effective site safety monitoring.

Table 10
Improved F1 scores by adding the weighted-class strategy for PPE classification

Model	Weighted-class	PPE classes				
		W	WH	WV	WHV	Average
VGG16	×	50.15	86.33	66.18	93.38	74.01
	√	72.22	91.36	89.25	96.48	**87.33**
ResNet50	×	47.11	96.22	55.26	97.45	74.01
	√	59.33	93.46	67.49	95.36	**78.91**

4.4. Applicative case of safety compliance monitoring

By combining the results of identity matching and PPE classification, the safety compliance of each worker is monitored with the developed software. In particular, incident reporting of workers not wearing the necessary PPEs is conducted with the DSD site videos. To this end, video clips spanning 1 month are processed end-to-end by the developed framework in Section 3, including worker detection, ReID, PPE classification and finally incident counting. For the two time thresholds T_b and T_r, incident reports under different values are generated for comparison, i.e. {0, 1, 3, 5} minutes which were suggested by the site officers of the DSD project. The on-duty hours were set as 9 am-5 pm excluding 12-1 pm, within which any missing-PPE incidents are logged and aggregated into a monthly report.

To exemplify the process of safety compliance monitoring, Fig. 14 illustrates some captured video screenshots of the workers, each associated with the ID index, classified PPE status and timestamp. Correspondingly, Fig. 15 shows the incident reports under different time buffers (T_b = 0, 1, 3, 5 min), summarizing the recorded counts of not wearing each type of PPE. For example, in Fig. 14, ID 1 does not wear helmet for a few frames (less than one minute), while he wears it back



Fig. 14. Samples of video screenshots with tracked IDs and PPE statuses for manual validation.

ID	WH (No vest)	WV (No helmet)	W (No helmet & vest)	Total	Timestamps
Total	0	0	0	0	

a) Incident report with $T_b = 5$ minutes

ID	WH (No vest)	WV (No helmet)	W (No helmet & vest)	Total	Timestamps
Total	0	0	0	0	

b) Incident report with $T_b = 3$ minutes

ID	WH (No vest)	WV (No helmet)	W (No helmet & vest)	Total	Timestamps
Total	0	0	0	0	

c) Incident report with $T_b = 1$ minute

ID	WH (No vest)	WV (No helmet)	W (No helmet & vest)	Total	Timestamps
1	0	1	0	1	['20210620_093255']
2	1	0	0	1	['20210706_150209']
Total	1	1	0	2	

d) Incident report with $T_b = 0$ minute

Fig. 15. Samples of incident report (based on Fig. 14) under different buffer time T_b.

very soon. Consequently, in Fig. 15(d), ID 1 is counted towards one "No Helmet" incident when no time buffer is considered (i.e. $T_b = 0$ min for reporting incident immediately). Yet, when considering time buffers (e.g. $T_b = 1$, 3, 5 min), ID 1 is not penalized since the duration of missing a helmet does not exceed the thresholds. The same interpretation applies to ID 2, who misses a vest only for a few seconds.

In practice, the worker images are captured for manual authentication of any incident reported, e.g. by safety officers. The output in Fig. 14 can be manually double-checked against the reports in Fig. 15, in order to validate and finalized the incident reports. By mapping the incident statistics with the saved images, the incidents were not considered as violating site safety since the worker took off and adjusted the PPEs only for a few seconds, in safe surroundings. In that case, it is reasonable to consider the worker as complying the PPE requirement.

As introduced in Section 4.1.3, the FDR under different T_b is reported to inspect the parameter influence on the incident reporting in practice. Hence, the proposed system is implemented end-to-end to process 1-h videos from the four cameras on the DSD site. According to Sections 4.2 and 4.3, the best models for worker ReID and PPE classification are used, i.e. ResNet50 and VGG16 respectively. Fig. 16 shows the computed FDRs under different buffer time T_b in seconds. The FDR generally drops when using a longer buffer time. When $T_b = 0$ (i.e. reporting incidents immediately), the FDR is at the highest value of 15%. Since under this setting the incident reporting is frame-by-frame, the falsely reported incidents are directly reflected from incorrect PPE classification results, i.e. a PPE-compliant worker on an image is wrongly classified as missing-PPEs. Since the PPE classification model achieves 87% F1 score, inaccurate PPE statuses may be produced which contribute to the FDR of incident reporting. Hence, as proposed in Section 3.4, a buffer time is necessary to tolerate imprecise PPE statuses to a certain extent.

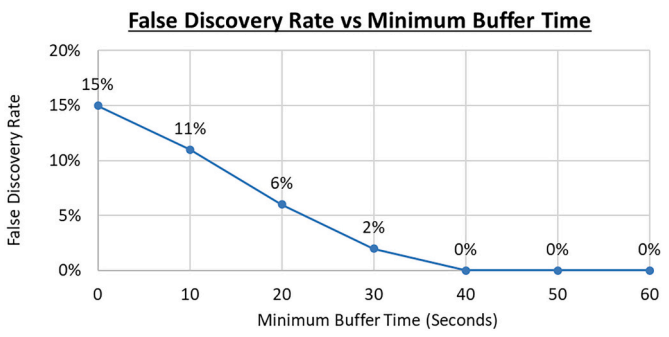


Fig. 16. False Discovery Rates for incident reporting under different time thresholds

From the same graph, the FDR steadily decreases as a longer buffer time is used, with only 2% when $T_b = 30$ s. Intuitively, introducing a buffer time effectively reduce the sensitivity of reporting incidents, since the missing-PPEs status of a worker needs to sustain a certain period to trigger the reporting. This notably mitigates the influence of misclassified PPE statuses, granting certain allowance for such error. This reveals the importance of introducing a time buffer to avoid recording an excessive number of falsely reported incidents. Capturing the screenshots for manual validation is also vital, which makes the incident reporting more transparent and justified. Furthermore, the FDR becomes 0% under $T_b \geq$ 40 s, meaning that the maximum duration of workers missing the PPEs is at most 40 s. Theoretically, such condition denotes the optimal value of T_b, with the lowest FDR obtained. Further enlarging the buffer time may report no any incident, which cannot reflect the precision of incident reporting. In practice, the choice of T_b should be site-specific and determined by the safety officers concerned.

4.5. Discussion of practical implications

Compared with previous studies which only tracked worker movement within a small area, performing worker ReID across multiple cameras contributes to a more practical monitoring of site safety compliance. With the ReID results, the identities of individual workers are consistently maintained within the monitoring area. The same worker is continuously tracked without identity switching, even when the worker leaves and enters different camera views at arbitrary timestamps, or when occluded by surrounding obstacles for a while and being detected again later. On the contrary, the existing single-camera analytics cannot maintain the same identities when workers walk

across different cameras in a large site area. This would make the incident counting inaccurate when considering a minimum time buffer before recognizing the incompliance of a worker. The worker may move around the whole site, such that the duration appearing on one camera may be shorter than the pre-defined time buffer. Without a consistent identity label, the observation timestamps cannot be accumulated, i.e. the state tracker is re-initialized whenever the worker moves to different cameras. Then, the number of incidents may be under-estimated since the same worker may miss any PPE throughout the multi-camera journey, which may be ignored since the timestamp counter is not maintained. With the ReID technique, the status of each worker can be tracked continuously for more accurate incident counting.

Several practical concerns upon the actual deployment of the monitoring system are considered, including the influence on construction site operations, system setup cost and computational efficiency when handling multiple cameras. Firstly, deploying the monitoring system will impose minimal influence on the construction site operations nearby. Literature suggested that installing multiple surveillance cameras on construction sites are fairly common [74], e.g. for safety officers to monitor onsite progress and crime security. Hence, the influence on the site operations should have been addressed upon camera placement, e.g. mounting at certain height and distance to capture videos from the top [75]. Such far-field camera setup is reasonable since it maximizes the utilization of the field of view of each camera [8], and enables site monitoring without disrupting the on-site activities [76]. Quite some studies of construction site monitoring are based on far-field cameras [61,77], whose feasibility of analyzing worker movement and behaviors have been validated.

Secondly, setting up the monitoring system will be economical. As mentioned, construction sites typically have surveillance cameras installed even though no computer vision engine is deployed. Since the proposed system is a generic workflow to process camera videos, the software can be readily integrated into the cameras already available on sites. In addition, the number of camera can be strategically optimized for cost-effectiveness. Several studies proposed systematic framework for camera layout design, e.g. by optimizing the field of view of each camera [78]. These could be referenced in actual deployment, e.g. only picking certain cameras on the optimal positions with maximized utilization of camera coverage. This will result in minimized number of cameras to be processed. Hence, the hardware and software setup are considered cost-effective.

Furthermore, the proposed monitoring system has a high computational efficiency in video processing, which can potentially enable real-time onsite alert. Table 11 summarizes the inference speeds of the system regarding its different stages, in frames per second (FPS). The worker detection stage is the slowest (50 FPS), mainly because it processes the raw video frames with large resolutions. Hence, the worker ReID and PPE classification stages which process the cropped worker images are reasonably faster (160 FPS and 200 FPS respectively). Even there are multiple workers in one frame, the cropped worker images are processed in parallel by the GPU in a batch, resulting in very fast inference compared with worker detection. Note that the incident reporting stage immediately returns the results with negligible delay (in milliseconds), mainly because it only processes numerical metadata (frame timestamps, worker identities and PPE classes), hence its inference time is omitted. Overall, the entire system can perform video inference in 32 FPS, which satisfactorily fulfills the 5-FPS requirement for real-time processing [10,18].

Table 11
Inference speeds for different stages of the proposed monitoring system

Stage	Inference speed (frames per second)
Worker detection	50
Worker ReID (identity matching)	160
Worker PPE classification	200
Entire system	32

The computational efficiency of the monitoring system can be further enhanced by designing parallel/distributed computational architectures, such as multithreading scheduling mechanism [79] and edge computing paradigm [80]. A computationally efficient framework potentially enables more proactive response to any unsafe behavior on sites (e.g. immediate alerts to not wearing PPEs or potential collisions with moving equipment). These techniques can enhance the scalability of the system for handling large number of cameras.

5. Conclusions and future work

The highly dynamic operations among workers and equipment have rendered construction sites hazardous, where the high on-site accident and fatality rates necessitate more effective measures of site safety monitoring. While a number of vision-based monitoring methods are available (e.g. to automatically track the movement of workers and identify their PPE statuses), this paper contributes over existing studies in the follow aspects:

1) A vision-based framework is proposed to analyse any unsafe behaviors of workers for safety monitoring on construction sites, specifically focusing on the compliance of wearing PPEs. This paper combines deep learning-based worker ReID and PPE classification to continuously track workers across multiple cameras and their PPE statuses over a site, enabling more comprehensive behavioral analyses among workers.
2) For worker ReID, a new loss function named similarity loss is designed to facilitate the learning of discriminative human features, to enable more robust tracking of individual workers. The proposed similarity loss notably improves existing loss functions (e.g. softmax and hard triplet losses) in training different deep learning models to extract more discriminative human features, with the IDF1 accuracy improved by 4% in two ReID datasets capturing different scenarios and human characteristics.
3) For PPE classification, a weighted-class strategy is designed to mitigate model bias when given imbalanced samples among classes, for improved performance towards the minority classes despite limited training samples. The developed strategy effectively encourages different deep learning models to learn robust features from the limited images of minority classes, with an improvement of 13% F1 accuracy in a construction site dataset. It would add practical values to identifying unsafe behavior among workers, of which training images are much scarcer than the compliant ones.
4) Combining the worker ReID and PPE classification results, a workflow is developed to log any incident of individual workers not wearing the necessary PPEs, for analyzing the safety compliance of workers over the whole site. Two time thresholds are defined for incident counting, which are adjustable to account for different habits of wearing PPEs among workers and produce more reasonable incident counts.

In the future, the proposed approach can be further enriched in by considering a wider range of on-site activities. Currently, the proposed framework mainly captures the movement and PPE statuses of construction workers for safety compliance monitoring, while more fine-grained categorization of their activities may enable richer behavioral analyses. For example, potential collisions with nearby equipment like trucks can be identified by monitoring the worker-equipment proximity on construction sites [16]. Besides safety, recognizing the task-specific actions of individual workers (e.g. undergoing masonry or plastering works) and tracing the duration spent on each task can support productivity evaluation [32].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the Drainage Service Department of the Hong Kong government for providing partial support to this research, in the project titled Construction of Dry Weather Flow Interceptor at Cherry Street Box Culvert (No. DC-2017-01). Some of the experimental data used in this research were collected from the construction site of the project. The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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